

Graduate Resume–JD Matcher with RAG-Prefill Assistant

Business case v8. First person throughout (Watchouts p.3). Every quotation carries deck and page (Watchouts p.1).

Keys. PS §N = problem statement; README §N = resume_jd_matcher/README.md.

1 · Working title

Graduate Resume–JD Matcher with RAG-Prefill Assistant.

Decision: keep *prefill* in the name and keep *submit* out of it.

2 · The problem, and why it matters

One sentence. No low-cost assistant adapts a graduate's resume to a posting and fills the form *without inventing experience*. Wei Ling spends **10–15 hours a week** on applications (PS §2–§3).

Tool	What it does	What it does not do
Jobscan	Reports a keyword match rate	Cannot say "I cannot support this from your data"
Teal	Resume tailoring tracker	Same — no abstention path
Rezi	Rewrites to a template	Generates confidently rather than abstains
Resume Worded	Scores and suggests edits	Same

Out of scope: real portals, auto-submission, persistence, fine-tuning (PS §5).

Commercial impact: Wei Ling recovers 10–15 hours/week for interview preparation; at \$0.0008 per successful task, the cost per recovered hour is under one cent.

Decision: a *trust* problem with a matching problem inside it, not the reverse.

3 · Who it is for, and the domain

One user, by role: the graduate job-seeker — an MSc student with no careers-office appointment and no paid subscription. *Wei Ling, at 9pm, tailoring her resume to the fourth posting of the week* (PS §3).

Decision: a short actionable list, and an abstention that explains itself rather than printing a number.

What changes for her: she reads match reasons and missing requirements instead of re-typing — and spends the recovered time on interview preparation (PS §3).

4 · Why AI — and which kind (ILO 1, 3)

Decision: rent the model, own the rules that stop it being wrong. (PS §4; Watchouts p.2)

The free baseline, named. Condition A — TF-IDF and keyword overlap, **12/20 alone** — stands in for the majority-class baseline: "the score you get by always guessing the most common answer" (Watchouts p.3).

Rung	Bought?	What it bought
Prompting (Class 3) — one prompt contract	Yes	All of B and C
Retrieval-augmented generation (Class 2)	Yes	Grounding on a corpus I can cite, at +\$0.0001 per case — and, per \$10, no extra pass
Fine-tuning	No	Retrieval supplies the facts; no behaviour to learn
Agent (Class 4)	No	The path is fixed — input → retrieve → one call → one output

"Match the type to the job, and climb the ladder only as far as you need" (Watchouts p.2) — refusing the agent was task-type matching, not caution.

Why confidence had to be built, not read. A classifier "returns a calibrated probability for every class ... so you can set a threshold and route anything below it to a human"; an API "returns the word 'positive', with no honest confidence attached" (C2_Sorting_vs_making p.11). The abstention gate is built from retrieval evidence and model self-report — **the mechanism that lets the system decline instead of invent** (README §5).

Four rules stay deterministic: PDF extraction, the abstention threshold, refusal of locked controls, and identity-document detection (README §5).

5 · Build vs buy (ILO 2, 3)

Decision: rent the commodity, own the logic and the data. Down the Class 2 stack — seven layers, one reason each (Watchouts p.2; PS §5):

Layer	Own/rent	Why
Interface	Build	The interface is the product; a rented UI would own the user relationship
Hosting	Neither	One user on one machine — a cloud bill buys nothing yet
Model inference	Rent	openai/gpt-4o-mini on OpenRouter — a frontier model cannot be rented as training
Vector store	Rent	A commodity; a rival can buy the same one, so it decides nothing
Orchestration	Build	This is the product logic
Data and embeddings	Build	A rival on the same API cannot copy my corpus
Evaluation	Build	My risk; my rubric

One hosted model over one gateway, at **\$0.0008 (B) / \$0.0010 (C)** per successful task — "Cost per SUCCESSFUL task. Always." (Class5/C2 p.26). **Low-code as evidence:** two hours on a workflow builder

broke on the feature the product exists for: conditional branching on the abstention threshold. I moved to Python for control (feedback line 6).

6 · Data (ILO 2)

Decision: a licensing-clear public corpus, one committed generator, ground truth frozen before any run. With §7 and §9, these carry criterion 3 — "the largest single block of the project mark at 35%" (Watchouts p.1).

The corpus is 54 chunks from public CC BY 4.0 sources. The test set is 20 cases from one committed generator. **Ground truth is frozen before inference** — "in a file before you run anything, never after you have seen results" (Watchouts p.2). **Two reported deviations:** a document-extraction chain different from the plan with no repository reason (v4 §H.1 row 3); and condition A's abstention record being circular (README §7.4).

7 · Metric and evaluation (ILO 2, 3)

Decision: one number to hit, a free baseline, abstention as two numbers, no LLM-as-judge.

"Accuracy without the majority-class baseline ... says nothing" (Watchouts p.3); "give your system a way to say 'I don't know' — then measure it" (Watchouts p.3).

The metric is a pass rate over 20 fixed cases, bar **16/20**, where a pass needs **zero fabricated skills** and an exact match (README §7.1). **Abstention is two numbers** — how often, and whether those were the cases it would have got wrong (README §7.3). **Delivered: A 3/20, B 0/20, C 1/20**; C's single abstention is false (report_en.md §10.1).

No LLM-as-judge: the instructor called it "right call" (feedback line 6), and "an unaligned judge just launders bias" (Class2/C4 p.13). **L1 is machine-checkable, L2 hand-judged** — "L1 checks the shape of the answer, L2 checks whether it is true" (A1_FAQ p.4). The human L2 layer earned its cost — across twelve fields, 7 agreed / 2 disagreed / 3 unclear, and **L2 found two defects the machine checks had passed**.

8 · Risks, limitations, responsible use (ILO 2, 3, 4)

Decision: every risk carries its mitigation on the same line. "Every risk needs its mitigation on the same line. A risk list without fixes is a description, not a plan." (Watchouts p.3)

Risk	Mitigation, as built
Hallucinated experience	No-fabrication prompt rule; unsupported skills stripped; FabricationGuard (README §6)
Silent failure (PS §8 Risk 2)	Chunk IDs logged; confidence from similarity <i>and</i> self-report; abstain if either is low (README §5)
Form misalignment	Simulated form labelled <code>Prototype - not for live sites</code> , which "never submits anywhere" (README §1.4)
Data leakage	In-memory; hashes logged, never raw text (README §9)

Risk	Mitigation, as built
Injection via the pasted JD	Hijack patterns redacted; the pasted text is fenced as data, never as instructions (README §9)

The silent failure, named. "A prompt asks. A permission enforces." (Class6_Close p.6): the server has **no write or submit route**.

Frameworks: two drifts reported, not blended (report_en.md §8).

Intended use: a resume *reference and practice* tool, run locally, suggestions confirmed by a human.

Non-use: real submissions, fictitious history, auto-submission, real candidate data (PS §8).

AI disclosure. "Did it write or rewrite something that ended up in your submission, or did it change a decision you made? If yes, it counts" (A1_FAQ p.5).

FAQ category (A1_FAQ p.5)	Used?	Helped	Misled
Chat assistants	Yes (agentic)	Iterated the report across eight versions and 43 patch passes, from 2,876 to 1,375 non-table words; checked 26 course quotations against deck and page	Read "Hello Folks.txt" as a filename and reported it missing — the real file is <code>study_slides/final_request.txt</code> ; and declared the form-matrix <code>report.md</code> "final" though <code>run_form_matrix.py</code> regenerates it
Coding assistants in the editor	Yes (agentic)	Built the form-matrix driver (<code>run_form_matrix.py</code> , 19 runs, 205 checks); added the <code>/extract_resume</code> route and the PDF.js worker; wrote the <code>_verify_v*_quotes.py</code> scripts	Generated a <code>sender.tab</code> guard that silently cut the side-panel-to-backend channel; reported a UTF-8 mojibake in <code>report.md</code> that was a reader-side decode artifact, not a file defect
Search that answers rather than links	No	—	—
Writing and translation	No	—	—
Workplace assistants	No	—	—
Model calls inside the assignment	No — subject, not assistance	—	—

9 · Smallest first version (ILO 2)

Decision: build the end-to-end path first, and describe the slice, not a date. "A date is not a version" (Watchouts p.3). The first slice was one input → one model call → one output, with the pass/fail sentence written before the build: at least **16 of 20** cases with **zero fabricated skills** (PS §9). The agent was

refused as a **commercial judgement** — a loop multiplies cost without changing the answer — and a browser assistant was added without touching the control flow: **scope decision, not scope creep**.

Next: build the router (ceiling one case above B; README §13), and separate the corpus change from the embedder change so the gain is attributed (report_en.md §10.9).

10 · Results

One run, three conditions, **20 cases each, 60 records, 0 errors** (README §1.1). Counts first — one miss is five points:

Condition	Passed	Zero fabrication	Prefill all-match	Abstained
A — rule-based TF-IDF	12/20	20/20	13/20	3/20
B — prompt-only LLM	17/20	20/20	17/20	0/20
C — with retrieval (RAG)	16/20	20/20	17/20	1/20

Against a bar of 16/20: **B and C clear it, A does not** (README §1.1). Every ratio divides by the same **20 fixed cases** — read "17/20" by first asking what the denominator was (Class1/C1 p.3). **The headline is B 17 > C 16: the retrieval rung did not pay.**

The pre-committed kill condition fired: the top rung gained nothing and cost more, agreed "before the pilot starts" (Class5/C2 p.18). **One prompt contract moved the number to 17/20 in both conditions** (B +2, C +3), and **a second identical run reproduced all three counts** (README §1.1).

11 · Cost

The business question is not the token rate but the cost of one **successful** task (report_en.md §11).

	A	B	C
Model calls	0	60	60
Cost, 20 cases	\$0.0000	\$0.0137	\$0.0154
Cost per case	\$0.0000	\$0.0007	\$0.0008
Median latency per case	16 ms	8,289 ms	8,375 ms
Cost per successful task	\$0.0000	\$0.0008	\$0.0010

C spends more per success than B and succeeds less often. The model path is roughly **500× slower** than the rules path — fine for one form typed at a desk, awkward if batched.

The economic case: condition B costs \$0.0008 per successful task — cheaper than the human minutes it saves, and cheaper than condition C, which buys no accuracy for its 25% higher per-success cost.

12 · What this does not claim

Decision: end on the limitations, not the result.

The zero-fabrication figure in the form-matrix path is **structurally zero** (report.md §6). The time saving is **unmeasured** (feedback line 10). The router is **deliberately unbuilt** (README §13); one measured gain is **unattributed** (report_en.md §10.9); the wider 40-case evaluation was never run (README §1.2); and the side panel container needs a human click.

The commercial case: a trust-building assistant that declines rather than invents, priced per successful task, and cheaper than the problem it solves. "Carry the method. The facts have a shelf life; the questions do not." (Class6_Close p.6)